



UNIVERSITY^{AT}ALBANY
STATE UNIVERSITY OF NEW YORK

CSI 436/536 (Fall 2026)
Machine Learning

Lecture 2: Review of Linear Algebra

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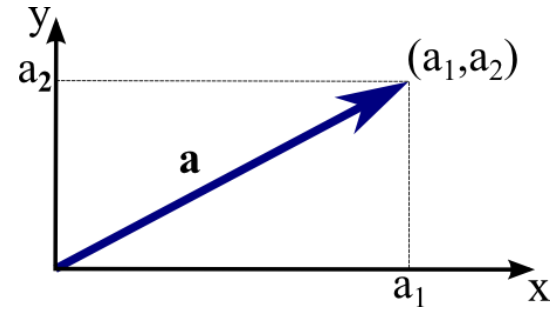
Aug 25, 2026

Today's agenda

- Key objects:
 - Vector, matrix
- Operations:
 - Matrix-vector multiplication, matrix-matrix multiplication
- Properties of vectors:
 - Norm (one vector), distance and angle (two vectors), linear (in)dependence, orthogonality (a “bag” of vectors)
- Properties of a matrix:
 - Rank, trace, determinant, symmetric, invertible

Vector and matrix

- Geometric meaning of a vector:
 - An arrow pointing from 0
 - A point in a coordinate system
- Matrix is a “bag” of vectors.
 - n-column vectors or m-row vectors.



$$\mathbf{a} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \in \mathbb{R}^3$$

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}, \quad a_{ij} \in \mathbb{R}.$$

Norms are “metrics”. A few useful properties:

Generally, a vector norm is a mapping $\mathbb{R}^n \rightarrow \mathbb{R}$, with the properties

- $\|x\| \geq 0$, for all x
- $\|x\| = 0$, if and only if $x = 0$
- $\|\alpha x\| = |\alpha| \|x\|$, $\alpha \in \mathbb{R}$
- $\|x + y\| \leq \|x\| + \|y\|$, for all x and y

l_p -norm is the most used vector norm

- Definition:

$$\|\mathbf{x}\|_p := \left(\sum_{i=1}^n |x_i|^p \right)^{1/p}$$

- Different norms:

- When $p = 1$, l_1 -norm, Taxicab norm, Manhattan norm $\|\mathbf{x}\|_1 := \sum_{i=1}^n |x_i|$

- When $p = 2$, l_2 -norm, Euclidean norm, quadratic norm, square norm

- In literature, $\|x\|$ usually denotes Euclidean norm

$$\|\mathbf{x}\|_2 := \sqrt{x_1^2 + \cdots + x_n^2}$$

- When $p \rightarrow \infty$, l_∞ -norm

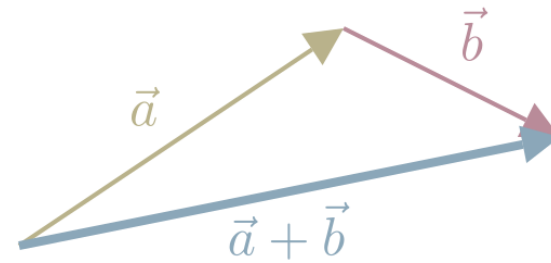
$$\|\mathbf{x}\|_\infty := \max_i |x_i|$$

In-class exercise

- Find l_1 -norm, l_2 -norm, l_∞ -norm of vector $x = [1, 2, 3, 4, -5]$.
- Answer: 15, $\sqrt{55}$, 5.

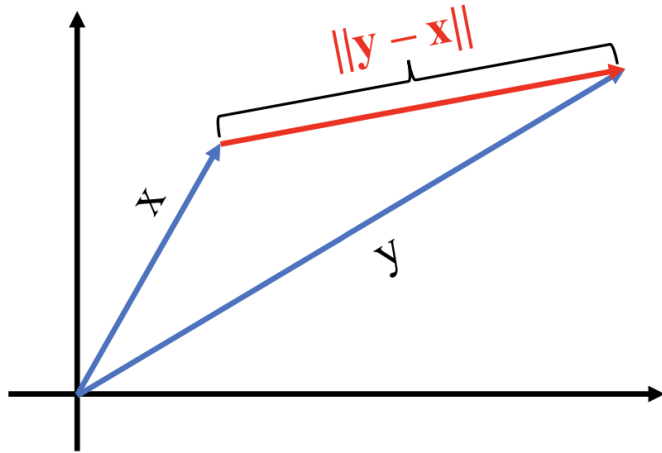
Properties of two vectors

- What can you do with them?
 - Add
 - $z = x + y$
 - $[5, 6, -2] = [1, 3, 5] + [4, 3, -7]$
 - Subtract
 - $g = x - y$
 - $[-3, 0, 12] = [1, 3, 5] - [4, 3, -7]$
 - Weighted combination / linear combination
 - $h = x + 2y$
 - $[9, 9, -9] = [1, 3, 5] + 2 * [4, 3, -7]$



Relationship (similarity) of two vectors

- Direction

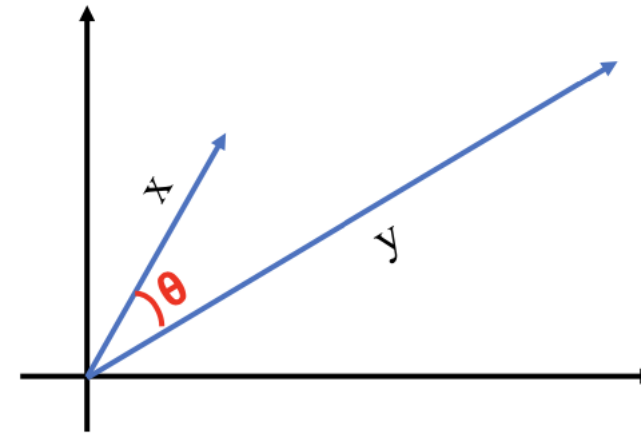


- Angle

- Dot product / inner product

- $\langle x, y \rangle = x^T y = \sum_{i=1}^n x_i y_i$

- $\theta = \cos^{-1} \left(\frac{\langle x, y \rangle}{\|x\| \|y\|} \right)$



Two vectors are **orthogonal** (perpendicular to each other) iff their dot-product is 0.

Three interpretations of matrix-vector Multiplication

- Interpretation 1: “Projecting x to m -directions”
 - Treat matrix A is as a “bag” of row-vectors
 - A is a m by n matrix
 - x is a n -dimensional vector
 - $Ax = \begin{bmatrix} 6 & 2 & 4 \\ -1 & 4 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ -2 \\ 1 \end{bmatrix} = \begin{bmatrix} 24 \\ -9 \end{bmatrix}$
 - Projecting x from 3 dimensions to 2 dimensions.

Three interpretations of Matrix-Vector Multiplication

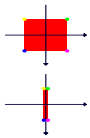
- Interpretation 2: “Weighted linear combination of column vectors”
 - Treat matrix A as a “bag” of column-vectors
 - A is a m by n matrix
 - x is a n -dimensional vector
 - $Ax = \begin{bmatrix} 6 & 2 & 4 \\ -1 & 4 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ -2 \\ 1 \end{bmatrix} = \begin{bmatrix} 24 \\ -9 \end{bmatrix}$
 - The weight of column 1 is 4
 - The weight of column 2 is -2
 - The weight of column 3 is 1

Three interpretations of matrix-vector Multiplication

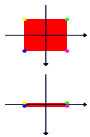
- Interpretation 3: “A linear transformation of input vector x ”
 - Treat matrix A as an “operator” or a “function that takes a vector input and output another vector” $A : \mathbb{R}^n \rightarrow \mathbb{R}^m$

Projection

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$



$$A = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

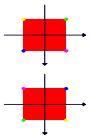


4 A projection onto a line containing unit vector $\vec{u}$ is $T(\vec{x}) = (\vec{x} \cdot \vec{u})\vec{u}$ with matrix $A = \begin{bmatrix} u_1 u_1 & u_2 u_1 \\ u_1 u_2 & u_2 u_2 \end{bmatrix}$.

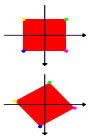
Projections are also important in statistics. Projections are not invertible except if we project onto the entire space. Projections also have the property that $P^2 = P$. If we do it twice, it is the same transformation. If we combine a projection with a dilation, we get a **rotation** dilation.

Rotation

$$A = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$



$$\hat{A} = \begin{bmatrix} \cos(\alpha) & -\sin(\alpha) \\ \sin(\alpha) & \cos(\alpha) \end{bmatrix}$$

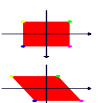


Any rotation has the form of the matrix to the right. Rotations are examples of orthogonal transformations. If we combine a rotation with a dilation,

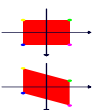
Shear

Shear transformations

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$



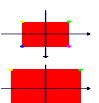
$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$



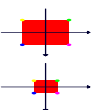
In general, shears are transformation in the plane with the property that there is a vector $\vec{w}$ such that $T(\vec{w}) = \vec{w}$ and $T(\vec{x}) - \vec{x}$ is a multiple of $\vec{w}$ for all $\vec{x}$. Shear transformations are invertible, and are important in general because they are examples which can not be diagonalized.

Scaling transformations

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$



$$A = \begin{bmatrix} 1/2 & 0 \\ 0 & 1/2 \end{bmatrix}$$



One can also look at transformations which scale x differently than y and where A is a diagonal matrix. Scaling transformations can also be written as $A = \lambda I_2$ where I_2 is the identity matrix.

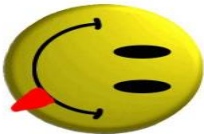
In-class exercise: map each pixel to a new location

A-F	image	A-F	image	A-F	image
					
					
					

b) The **smiley face** visible to the right is transformed with various linear transformations represented by matrices $A - F$. Find out which matrix does which transformation:

$$A = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix},$$

$$D = \begin{bmatrix} 1 & -1 \\ 0 & -1 \end{bmatrix}, \quad E = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}, \quad F = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} / 2$$



Solutions:

Suppose the **red** tongue pointing towards right-bottom is represented by a vector $x = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

Then

$$Ax = \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \quad Bx = \begin{bmatrix} -1 \\ -1 \end{bmatrix}, \quad Cx = \begin{bmatrix} 1 \\ 1 \end{bmatrix},$$

$$Dx = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad Ex = \begin{bmatrix} -1 \\ -1 \end{bmatrix}, \quad Fx = \begin{bmatrix} -0.5 \\ -0.5 \end{bmatrix}$$

Final solutions are:

$$\begin{bmatrix} D & C & E \\ A & B & F \end{bmatrix}$$

Matrix-Matrix multiplication

- Let $A \in \mathbb{R}^{m \times p}$ and $B \in \mathbb{R}^{p \times n}$. Then,
 $C = AB = (c_{ij}) \in \mathbb{R}^{m \times n}$ is defined as follows:

$$c_{ij} = \sum_{k=1}^p a_{ik} b_{kj}, \text{ for all } i = 1, \dots, m, j = 1, \dots, n.$$

- Key things to remember
 - Dimension check!
- Properties of a scalar-scalar multiplications (which ones are still valid for matrix-matrix multiplication?)
 - Commutative law: $AB=BA$?
 - Associative law: $(AB)C=A(BC)$?
 - Distributive law: $A(B+C)=AB+AC$?

Examples of matrix-matrix multiplication

- Inner product and outer product of two vectors

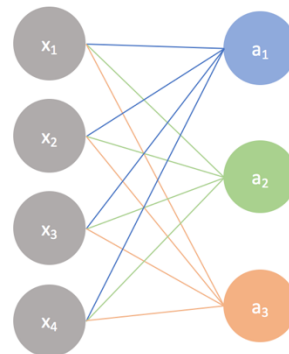
$$\mathbf{u} \otimes \mathbf{v} = \mathbf{u}\mathbf{v}^T = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix} \begin{bmatrix} v_1 & v_2 & v_3 \end{bmatrix} = \begin{bmatrix} u_1 v_1 & u_1 v_2 & u_1 v_3 \\ u_2 v_1 & u_2 v_2 & u_2 v_3 \\ u_3 v_1 & u_3 v_2 & u_3 v_3 \\ u_4 v_1 & u_4 v_2 & u_4 v_3 \end{bmatrix}$$

- Page rank (mathematics behind Google Search)

- <https://pi.math.cornell.edu/~mec/Winter2009/RalucaRemus/Lecture3/lecture3.html>

- Neural networks

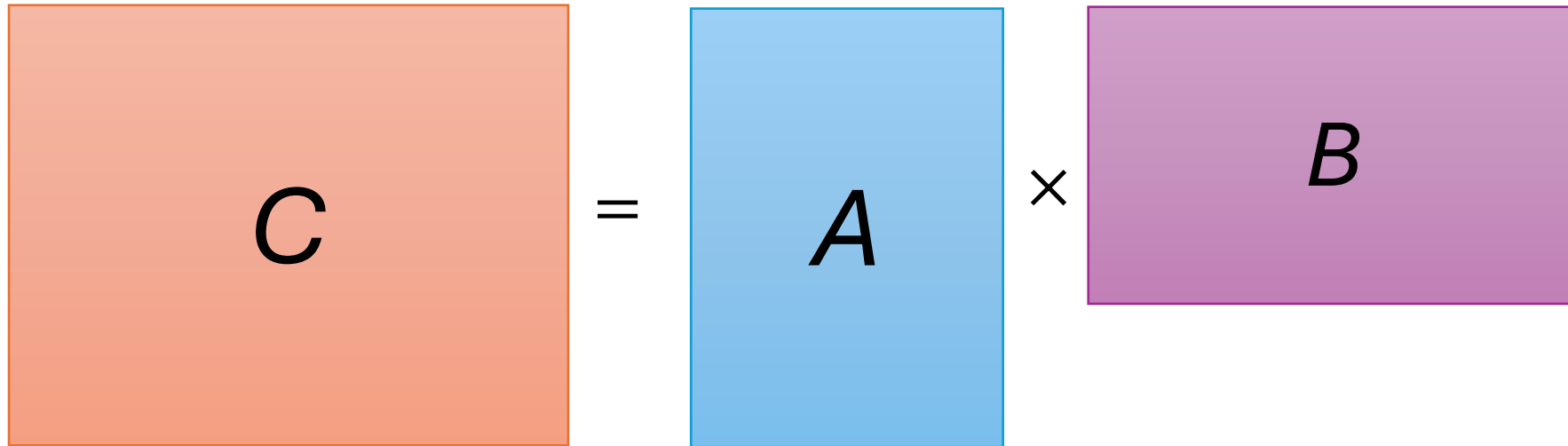
Input layer Output layer



A simple neural network

$$\begin{bmatrix} w_1 & w_2 & w_3 & w_4 \\ w_1 & w_2 & w_3 & w_4 \\ w_1 & w_2 & w_3 & w_4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} b \\ b \\ b \end{bmatrix} = \begin{bmatrix} w_1 x_1 + w_2 x_2 + w_3 x_3 + w_4 x_4 + b \\ w_1 x_1 + w_2 x_2 + w_3 x_3 + w_4 x_4 + b \\ w_1 x_1 + w_2 x_2 + w_3 x_3 + w_4 x_4 + b \end{bmatrix} \xrightarrow{\text{activation}} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$$

Computational Complexity of matrix Multiplication



A diagram illustrating matrix multiplication. It shows three colored rectangles: an orange square labeled C , a blue rectangle labeled A , and a purple rectangle labeled B . The orange square C is on the left, followed by an equals sign $=$, then the blue rectangle A , followed by a multiplication symbol $\times$, and finally the purple rectangle B . This represents the equation $C = A \times B$.

- In-class exercise: Suppose A is $m \times n$ and B is $n \times p$.
 - Step 1: How many dot products needed?
 - Step 2: How many multiplications in each dot product?
 - Step 3: Computational complexity of matrix multiplication?

Fun fact: complexity of matrix multiplication is still an open problem

- Multiplication of two 2 by 2 matrices
 - Naïve algorithm takes 8 multiplication
 - **Strassen** showed that one can get away with 7
- Divide and conquer gives
$$O(n^{\log_2 7}) \approx O(n^{2.807})$$
 - Improves over $O(n^3)$ for reasonable sized matrices
- Actually used in practice!

Timeline of matrix multiplication exponent

Year	Bound on ω	Authors
1969	2.8074	Strassen ^[1]
1978	2.796	Pan ^[9]
1979	2.780	Bini, Capovani ^[it] , Romani ^[10]
1981	2.522	Schönhage ^[11]
1981	2.517	Romani ^[12]
1981	2.496	Coppersmith, Winograd ^[13]
1986	2.479	Strassen ^[14]
1990	2.3755	Coppersmith, Winograd ^[15]
2010	2.3737	Stothers ^[16]
2012	2.3729	Williams ^{[17][18]}
2014	2.3728639	Le Gall ^[19]
2020	2.3728596	Alman, Williams ^{[20][21]}
2022	2.371866	Duan, Wu, Zhou ^[22]
2024	2.371552	Williams, Xu, Xu, and Zhou ^[23]
2024	2.371339	Alman, Duan, Williams, Xu, Xu, and Zhou ^[24]
2026	2.371177	Dupont, Eisenberger, Kozlovskii, Mehrabian, Ruiz, See, Zhou, Alman, Williams, Balog ^[2]

Best lower bound is still $\Omega(n^2 \log n)$

Properties of a bag of vectors: linear independence

Important to consider for machine learning algorithm design

- Given a set of vectors $\{v_1, v_2, \dots, v_n\} \in \mathbb{R}^m$, with $m \geq n$, consider the set of **linear combinations** $y = \sum_{j=1}^n \alpha_j v_j$ for arbitrary coefficients α_j 's.
- The vectors $\{v_1, v_2, \dots, v_n\}$ are **linearly independent**, if $\sum_{j=1}^n \alpha_j v_j = 0$, if and only if $\alpha_j = 0$ for all $j = 1, \dots, n$.
- Implication: if a set of vectors are linearly dependent, then one of them can be written as a linear combination of the others

In-class exercise: linear independence

Are these vectors linear dependent?

$$v_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, v_2 = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}, v_3 = \begin{pmatrix} 3 \\ 1 \\ 4 \end{pmatrix}$$

Yes, because that $2v_1 + v_2 - v_3 = 0$. Or equivalently,
 $v_3 = 2v_1 + v_2$.